Mathan Prasanna Kumar S

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Experience

Software Developer (Vision Systems), RobotoAI Technologies Jan 2024 – Nov 2025

- Led and developed a real-time production monitoring system to track products on a conveyor line using YOLO and ByteTrack for detection and tracking, OCR for label extraction, SQLite for data logging, and collaborated with the backend team for live dashboard integration
- Integrated visual odometry and IMU from a stereo camera into the navigation stack to enhance localization and navigation of an autonomous mobile robot, and developed multithreaded ROS2 nodes for efficient communication between hardware components
- Collaborated with the navigation team to enable vision-based obstacle avoidance to enhance navigation robustness and safety
- Developed an autonomous docking system using ArUco markers by estimating pose with OpenCV, transforming coordinates to ROS frames, and integrating with the Nav2 docking server for precise alignment
- Implemented a person following feature where optimized detection models using TensorRT for low-latency inference, and developed a motion controller for smooth and responsive movement
- Developed a pressure-based altitude measurement system on MSP430, handling SPI and UART communication, real-time data acquisition, filtering raw data to compute accurate altitude
- Led the development of Voice Assistant for AMR using AI agents, enabling the robot to understand and execute spoken instructions

Software Developer Intern, RobotoAI Technologies Sep 2023 – Dec 2023

- Researched and developed a BLE based localization system for improving the global localization in dynamic environments. Handled data transmission from beacons to gateway through MQTT protocol, analyzed and applied low low-pass filter to the raw RSSI data, and developed a triangulation algorithm to find the gateway position
- Developed an A* path planning algorithm for the line follower robot to find the shortest trajectory between stations
- Implemented a peer-to-peer video streaming application by using the Gstreamer WebRTC plugin, enabling the user to teleoperate the robot at remote locations
- Developed a chip detection application to distinguish between the threaded and unthreaded chips. Used YOLO model and implemented the training and inference pipeline in Pytorch, and further optimized the model for accelerated inference in TensorRT

Skills

Languages: C, C++, Python, Javascript, SQL
Frameworks: Pytorch, OpenCV, ROS2, TensorRT, LangGraph
Tools: Docker, Git, CMake

Projects

T5_Torch mathanprasannakumar/T5_Torch

- Implemented the T5 transformer architecture from scratch in PyTorch to deepen the understanding of attention mechanisms and the encoder-decoder structure
- Integrated training, inference, and dataloader pipelines, and loaded pretrained weights from HuggingFace to validate the performance and enable fine-tuning on custom tasks

Local_RAG mathanprasannakumar/Local_RAG

- Built a Streamlit application that enables users to upload a PDF and ask context-based questions. Used LangChain to process and chunk documents, generate embeddings, and store them in a Chroma vector database, and augmented the query with top-matched chunks to provide grounded responses via an LLM

Education

Bachelor of Engineering in Civil Engineering April 2022

Government College of Engineering, Salem, TN